Daniel Vargas
Norman, OK 73069 | dvarg88@gmail.com | linkedin.com/in/daniel-vargas-slydly/ | dannyboy849.github.io

EDUCATION

University of Oklahoma, Norman, OK — Expected May 2028
Ph.D. in Mechanical Engineering — Control Concentration, GPA: 3.67
Dissertation: Estimation and predictive control for drone teams without GNSS, using learned dynamics and decentralized state estimates with calibrated uncertainty; validated on a 5-drone fleet
B.S. in Mechanical Engineering, Minor: Electrical and Computer Engineering — May 2024

TECHNICAL SKILLS

Control & Optimization: Model Predictive Control (MPC/NMPC, real-time), Learning-Based Control, LQR, Trajectory Optimization, Numerical Optimization (QP, NLP, MILP), System Identification
Estimation and Perception: State Estimation (EKF/UKF), Sensor Fusion, SLAM/V-SLAM, LiDAR Odometry, Computer Vision, Object Tracking
Machine Learning: PyTorch, scikit-learn, Neural Network & Gaussian-Process Dynamics Models, CNNs, Hyperparameter Optimization (Optuna)
Robotics Software & Tools: ROS 2, ArduPilot/PX4, MAVLink, SITL, Gazebo, Isaac Sim, CasADi/IPOPT, do-mpc, OpenCV
Programming: Python, C/C++, MATLAB, CUDA, Linux, Git, Docker, acados

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Oklahoma, Norman, OK — Aug 2024 – Present

Selective Learned-Dynamics MPC for VTOL Aircraft (DATUM) — *JGCD, [submitted 2026]*

- Built a predictive controller that learns neural-network corrections only for states where a held-out 1-s prediction test shows learning helps; identified from 66 min of autopilot flight logs of a 21.5-kg lift-plus-cruise VTOL.
- On an unseen flight: lowest error in every wind scenario; matched or beat the best classical MPC (12% better than linear MPC on the landing) and ~30% lower error than a dense neural MPC, with 2–5× less variability; 5 seeds, 7 baselines."
- Cut solve time from ~300 ms to ~30 ms (10×) via per-step linearization, meeting a 10-Hz loop; validated against ArduPilot SITL in real time."
- Python, CasADi/IPOPT, do-mpc, scikit-learn, Optuna; reproducible pipeline with unit tests.

Comprehensive AI Risk Assessment (CARA) UAV Platform

- Designed a modular autonomous UAV framework for risk-aware mission planning and medical logistics.
- Developed graph-based route optimization, multi-commodity-flow hub placement, weather-aware planning, and data-driven MPC.
- MILP hub placement: 3–10% lower routing cost than distance-based siting; random/coverage heuristics infeasible in 98–100% of constrained trials
- Implemented ROS 2/ArduPilot architecture supporting waypoint generation, energy-aware trajectory planning, and real-time replanning.
- Developed mission-planning architecture for operation under environmental uncertainty.

UGV Risk-Aware Navigation in Construction Sites

- Developed predictive control architecture for autonomous UGV navigation in dynamic construction environments.
- Integrated learned human behavior models with path planning and control to account for latent safety hazards.
- Designed real-time autonomous navigation strategies for operation around humans and obstacles.

Data-Driven Steering Dynamics Modeling and Control

- Contributed to the software implementation, methodology, and experimental workflow for data-driven steering dynamics modeling and steering-angle tracking control.
- Supported development and evaluation of the vehicle modeling and control pipeline using experimental drive-by-wire data.

Drone-Based Methane Emission Detection

- Developing nonlinear MPC-based UAV pursuit control for autonomous gas-plume tracking.
- Integrated CNN-based plume segmentation with predictive trajectory generation.
- Investigating constrained autonomous flight under environmental uncertainty.

AirOU M-City: Autonomous Vehicle Testbed

- Led a 5-person team developing a ROS-integrated autonomous vehicle testbed with V2I communication for research and education.
- Integrated computer vision, LiDAR/VSLAM, autonomous navigation, and connected-vehicle infrastructure.
- Published at IEEE ITSC 2024.

Graduate Teaching Assistant

University of Oklahoma, Norman, OK — Jan 2025 - May 2026

- Instructed 120+ students in Python-based AI and machine-learning laboratories.

- Led 15+ applied ML/robotics sessions.

Undergraduate Research Assistant
University of Oklahoma, Norman, OK Jan 2022 – May 2024

- Designed and built autonomous robot, ARCPro, used in computer vision research and academic course.

PUBLICATIONS

Under Review:

- **Vargas, D**.; Habibi, G. "Learning Only What Matters: State-Selective Residual Models for Predictive Control." J. Guidance, Control, and Dynamics (submitted)

Published:

- **Vargas, D**. et al. "Design and Implementation of Smart Infrastructures and Connected Vehicles in a Mini-City Platform." *IEEE Intelligent Transportation Systems Conference (ITSC), 2024.*
- Simpore, F.; **Vargas, D.** et al. "Data-Driven Steering Dynamics Modeling and Steering Angle Tracking Control for Self-Driving Vehicles: Simulation and Experiments on 2025 Nissan Leaf Electric Vehicle" *Sensors.*
- Habibi, G.; Billerbeck, S.; **Vargas, D.** et al. "Development of a Hands-On Robotic Course on Computer Vision for Autonomous Vehicles." *IEEE ISEC, 2025.*
- Dalton, C.; Mubassira, S.; **Vargas, D.**; Haldavnekar, R. "Reverse Engineering the Pre-Capstone Experience: A Practice-Oriented Model for Industry-Relevant Engineering Education" *ASEE Annual Conference & Exposition, 2026.*